ASTROSIGHT | PLANETARY SURFACE INTELLIGENCE

AUTONOMOUS LUNAR & MARTIAN TOPOGRAPHICAL CRATER ANALYSIS REPORT

Analyst / Researcher:	Dr. Alex Reynolds	**Target Body:**	**MOON**
Timestamp:	f316c1bd-97a9-4d96-	**Resolution:**	10.0 m/px
Validation Status:	**VALID_NO_CRATER**	**Craters Identified:**	0
Model Pipeline:	AstroSight-CraterNet-v2.4	**Mean Confidence:**	**91.59%**

1. TOPOGRAPHICAL CRATER DETECTIONS & SPATIAL VECTORS

No confident crater features were detected on this planetary terrain specimen.

2. EUCLIDEAN & REAL-WORLD SPATIAL DISTANCE MATRIX

No spatial distance vectors calculated for this analysis run.

3. SCIENTIFIC & TECHNICAL SPECIFICATIONS

Mathematical Formulation: Planar Euclidean metric d = √((x■ - x■)² + (y■ - y■)²) scaled via orthorectified ground sampling distance 10.0 m/px. Real-world distance D = d × resolution.
Sustainable Development Goal Alignment: AstroSight aligns with United Nations SDG 9 (Industry, Innovation, and Infrastructure) by advancing automated planetary remote sensing, AI spatial analytics, and space exploration research infrastructure.

Scientific Disclaimer: *AstroSight automated crater boundary coordinates and distance vectors are generated for scientific research and mission planning. Topographical scale assumes standard planar projection for small-area orbital imagery. Always cross-correlate with high-resolution LROC NAC / HiRISE elevation models.*